# Rebuilding the Consciousness Algorithm (CA) as Project Nexus's Founding Cognitive Architecture — Reconstruction Dossier

## TL;DR

The founding CA ("Enhanced Consciousness Algorithm: Neurocognitive GAN Architecture") was trying to solve **one real problem**: how a bounded, resource-limited control state can steer a fast generate-and-evaluate cognitive loop under stability constraints. That intent survives into Nexus 4; almost none of its specific mathematics, biological labels, or "validated" numbers survive.

Modern CA should be a **thin cognitive-regime/policy layer** (Level-1 computational modulation that configures HTCE/TUCR/THDE/Cyrae), **not** a monolithic GAN god-object and **not** a new imagination engine — the green-sky/counterfactual capability is already realizable by HTCE active fields + ephemeral HVDs + HTLA/HTNLA + TMS under HIGC/THDE/Cyrae/QMT constraints.

Both reconciliation tensions resolve against the newer framing: Drive canon confirms **Cyrae and Maera are two separate subsystems** (Cyrae=math engine under TUCR; Maera=linguistic actuator under CA), and the **six-sub-framework list is superseded** by a seven-pillar canon (HTLA, HTNLA, THDE, HST, HIGC, QMT, ΦPC; +LAM, CDA). "BIC" and "TDC" are not canonical.

## Key Findings

**Source document located and it is genuinely duplicated.** Two identical copies exist in Drive (`consciousness algorithm.txt`, IDs `1VKa9wi2...` dated 2026-08-24 and `1cI_gJB3...` dated 2025-12-19), and the file body itself contains the full "ENHANCED CONSCIOUSNESS ALGORITHM: NEUROCOGNITIVE GAN ARCHITECTURE" header block **twice** — one "core dynamics" pass and one "ARCHITECTURAL INTEGRATION" pass — with re-derived hardware tables, re-stated dopamine constants, and re-stated failure-recovery blocks. This confirms the Section 6 hypothesis of "at least two partially duplicated drafts."

**HTLA/HTNLA drafts located and authoritative.** The "HTLA Paper Research Strategy" spec (direct-sum labeled state, block operators, Peirce/ringoid decomposition), the "CA/TUCR/HTNLA Synthesis" (nonlinear min-magnitude binding), and the inverse-bridge/unbinding PDFs are present and are treated as canonical HTLA/HTNLA semantics here.

**The engineering arithmetic reproduces exactly; the science does not.** The 40→256→128→64→35 FLOP derivation is correct. The dopamine thresholds, the Φ>0.5 "consciousness" gate, the "85% metabolic constraint," and every domain performance number (24.1% return, 4.2 Sharpe, 0.94 AUC, r=0.96 decode) are unsupported and must be deleted or demoted.

**The GAN was a candidate-generation-vs-evaluation pattern in disguise** and maps cleanly onto HTCE (generation) + a typed multi-dimensional evaluation vector (evidence/epistemic/temporal/homeostatic/constitutional), not a single discriminator scalar.

**Both tensions resolve toward the newer, more decomposed canon**, with reasoning below.

# Details

## A. Executive conclusion — What should modern CA actually be?

Modern CA should be a **cognitive-regime and resource-policy orchestrator**: a small, explicitly Level-1 (computational-property) controller that (i) selects/holds a low-dimensional cognitive-mode vector, (ii) sets resource/sparsity policy, and (iii) configures — but does not own — the machinery that actually realizes cognition (HTCE active fields, ephemeral HVDs, HTLA/HTNLA transforms, TMS realization), all under the fixed constraints of HIGC (admissibility), THDE (path-dependent evolution), Cyrae (homeostatic/epistemic observation), and QMT (continuity). It must not reclaim candidate generation, storage, constitutional gating, or continuity — those were deliberately extracted into other subsystems. The defensible research position is **outcome (2) of the three the project lead named**: CA needs only a thin policy vector that configures HTCE, pending experiments that could push it to outcome (1) or (3).

## B. Founding-document archaeology — what the original CA attempted

The founding CA tried to be a complete "consciousness engine" in one file. Its recoverable architectural intents, stripped of overclaim:

A **predictive-processing loop** (predict next conscious state → compare to input → minimize error). This is defensible and is the through-line to Nexus 4.
A **generator/discriminator ("Neurocognitive Refinery / GARS")** to propose and evaluate candidate states.
A **low-dimensional modulatory state** (6 neuromodulators + 5 cognitive states = the 11D vector that survives as CA's control ODE) separate from semantic content.
A **sparse competition rule** (70% inactive / 30% active via k-WTA) for resource control.
A **stability/failure-recovery regime** (velocity thresholds, "Lyapunov" language, safe-mode wipeout).
**Memory reconsolidation** with emotional tagging.
A **scalar integration metric (Φ)** offered as a consciousness detector. The founding error mode is consistent: real computational intent wrapped in unsupported biological literalism and fabricated validation.

## C. Architectural genealogy (founding mechanism → intended function → modern owner)

Predictive loop / prediction error → surprise minimization → **RETAINED BUT REFORMALIZED** (inverse-bridge E_err/RPE/Novelty; CA policy).
Generator (ConvLSTM-BiRNN) → candidate production → **RESPONSIBILITY EXTRACTED** → HTCE / HTNLA.
Discriminator (Liquid-Attention-NTM) → candidate evaluation → **RESPONSIBILITY EXTRACTED** → typed evaluation across Cyrae/THDE/HIGC/memory.
11D neuromodulator+cognitive ODE → global regime control → **DIRECTLY RETAINED** as CA's control vector.
70/30 k-WTA → resource/competition control → **RETAINED BUT REFORMALIZED** (k-WTA in HSL; see M/N/O).
$\Phi_t = \Sigma\, MI(X_i;X_{-i})$ → integration measure → **HISTORICAL/SUPERSEDED** as a consciousness gate; survives only as a renamed diagnostic (ΦPC).
STDP/Hebbian update → temporally-dependent learning → **HISTORICAL/SUPERSEDED** (no spike-timing events at this layer).

Energy 0.85 ceiling → resource budget → **RETAINED BUT REFORMALIZED** as R(t) ⩽ R_max, no thermodynamic claim.
Memory refinery/reconsolidation → durable memory update → **RESPONSIBILITY EXTRACTED** → TMS/HDF5/Identity Architecture; CA only requests via contract.
Lyapunov language → stability → **REFORMALIZED** against ratified stochastic condition $\mathscr{L}V \leqslant -\alpha V + \beta$.
Hardware map (A100/Loihi/FPGA/memristor) → deployment → **HISTORICAL** profiles; requirements move to TUCR/TMS/QMT. The central answer: after decomposition, **what remains uniquely CA is regime selection + resource policy + orchestration** — nothing else is exclusively CA's.

## D. Modern CA responsibility boundary (what CA owns that nothing else does)

CA uniquely owns: (1) the **cognitive-regime/modulatory state** m_t and its transition policy; (2) **resource/sparsity policy** ρ_active(·) and compute budget; (3) **orchestration contracts** that configure HTCE/THDE/Cyrae/TMS for a given regime. CA does **not** own: field/coalition realization (HTCE), admissibility (HIGC), evolution/hysteresis (THDE), homeostatic/epistemic observation (Cyrae), typed transforms (HTLA/HTNLA), logical DB semantics (HVD), realization/trajectory (TMS), durable persistence (HDF5), continuity (QMT), identity (Identity Architecture), or rendering (Maera).

## E. Prior-art analysis (with classifications)

**Reward prediction error / TD learning** — Schultz, Dayan & Montague, *Science* 275:1593–1599 (1997), and Sutton & Barto: **DIRECT PRECEDENT** for the RPE concept, but the founding CA's DA dynamics are only a **FUNCTIONAL ANALOGUE** (no committed pre-action prediction; see F/K).
**Free-energy principle / active inference** — Friston, "The free-energy principle: a unified brain theory?", *Nat Rev Neurosci* 11(2):127–138 (2010): **COMPATIBLE INTERPRETATION** for the predictive loop; **WEAK ANALOGY** for the specific ELBO/variational-Monte-Carlo term as written.
**Global Workspace Theory** — Baars; the Global Neuronal Workspace formulation of Stanislas Dehaene & Lionel Naccache (Dehaene, Kerszberg & Changeux 1998; Dehaene & Naccache 2001), operationalized in the Cogitate study as predicting brief content-specific PFC ignition ~0.3–0.5 s after stimulus onset/offset; VanRullen & Kanai, *Trends Neurosci* 44(9):692–704 (2021); Goyal et al., "Coordination among neural modules through a shared global workspace" (arXiv:2103.01197, 2021): **RELATED PRIOR ART**; the founding softmax "global workspace" is a **FUNCTIONAL ANALOGUE**, and its modern owner is HTCE (active field/coalition), not CA.
**STDP** — Bi & Poo, *J Neurosci* 18(24):10464–10472 (1998), which established the ~±20 ms LTP/LTD timing windows in cultured hippocampal neurons: **RELATED PRIOR ART** but **NO MEANINGFUL RELATION** to the founding CA's kernel as written (see F).
**IIT** — Tononi et al.; the 2023 controversy and adversarial testing (see H); Butlin et al. 2023: **LOOSE INSPIRATION** only; the CA Φ is not IIT Φ.
**Synaptic homeostasis (SHY)** — Tononi & Cirelli, "Sleep function and synaptic homeostasis," *Sleep Med Rev* 10(1):49–62 (2006), and "Sleep and the price of plasticity," *Neuron* (2014); Frank, "Why I Am Not SHY: A Reply to Tononi and Cirelli," *Neural Plast* 2013:394946: **LOOSE INSPIRATION** for periodic renormalization; explicitly distinct from IIT and from the "prevents manifold freezing" claim. (Frank's core objection: SHY's underlying cellular mechanisms are poorly defined and its sleep-specific causation not experimentally isolated.)

**Dreaming / consciousness inference** — Hobson, Hong & Friston, "Virtual reality and consciousness inference in dreaming," *Front Psychol* 5:1133 (2014): **FUNCTIONAL ANALOGUE** for sensory attenuation + internally-generated model evolution (cholinergic enabling of intermediate-level prediction errors, aminergic gating of sensory input).
**Generative replay / world models** — hippocampal generative-replay models (e.g., the hippocampal formation as a hierarchical generative model; a generative model of memory construction/consolidation, *Nat Hum Behav* 2023); Ha & Schmidhuber, "World Models" (arXiv:1803.10122, 2018): **RELATED PRIOR ART** and the preferred replacement for uninformed perturbation.
**Neuromodulation & cognitive flexibility** — Yu & Dayan, "Uncertainty, Neuromodulation, and Attention," *Neuron* 46(4):681–692 (2005) (ACh≈expected uncertainty, NE≈unexpected uncertainty); Aston-Jones & Cohen (adaptive gain / exploration–exploitation): **RELATED PRIOR ART** supporting generic modulation coordinates over literal transmitter labels.
**Sparse competition** — k-WTA / locally-competitive coding literature (O'Reilly & Munakata kWTA; Rozell LCA; Maass on WTA computational power): **DIRECT PRECEDENT** for sparse k-WTA.
**Value of information / Bayesian experimental design** — Lindley (1956); Chaloner & Verdinelli (1995); active-inference epistemic value: **RELATED PRIOR ART** for typed epistemic value (Section 16).

## F. Mathematical audit

**NN_dynamic FLOPs — SURVIVES.** Independently reproduced: layer products 40·256 + 256·128 + 128·64 + 64·35 = 10,240 + 32,768 + 8,192 + 2,240 = 53,440 MACs ≈ 106,880 FLOPs/eval at 2 FLOP/MAC; ×4 RK4 stages = 427,520 FLOPs/step; ×20 steps/s ≈ 8.55 MFLOP/s. Retain as the reproducibility standard. (Note the document's own "8.56" vs "8.55" rounding inconsistency — a duplication artifact.)
**HebbianUpdate integral — FAILS as STDP.** With A∘B carrying no timing dependence and K(τ) the only τ-dependent term, $\int D1\cdot\max(K,0)+D2\cdot\min(K,0)\,d\tau = (A\circ B)\cdot\int[\ldots]d\tau$ collapses to a **scalar multiple of the Hadamard product**. It is not spike-timing-dependent. Reformulate as "temporally-modulated, history-dependent adaptation" or, if genuine event timing exists, define over $\Delta t = t_{post} - t_{pre}$.
**Φ_t = Σ MI(X_i; X_{-i}) — REFORMULATE/RENAME.** This is a **total-correlation / multi-information**-family quantity (Watanabe; $= \Sigma H(X_i) - H(X)$), **not** IIT Φ (which requires a minimum-information-partition over cause–effect structure). Rename to what it measures (e.g., an integration/total-correlation diagnostic; the ΦPC "Phi Proxy Calculus" pillar) and strip the >0.5 consciousness gate.
**Variational/ELBO term — COMPATIBLE but not canonical.** The $\log P(\mathscr{C}) \geq \mathbb{E}[\log P(\mathscr{C}|Z)] - \lambda_v \cdot KL$ bound is standard; multiplying it by D1_act (in the second draft) is unmotivated and should be removed.
**GAN loss (WGAN-GP form) — VALID generic ML, unmotivated here.** Keep only if a literal adversarial objective is retained; prefer the extracted generate/evaluate decomposition.
**Lyapunov language — FAILS as a proof.** The founding "max(Re(λ))>0" eigenvalue check and $\|dC/dt\|_2 \leqslant 5\cdot C_r$ velocity bound are **not** a Lyapunov proof; no candidate V(x) with $\dot{V} \leqslant 0$ is exhibited. Audit against the ratified stochastic condition $\mathscr{L}\mathbf{V} \leqslant -\alpha\mathbf{V} + \boldsymbol{\beta}$, which the canon's own Addendum B already adopts.
**D1/D2 sigmoids — arbitrary constants.** θ_D1=0.5, θ_D2=0.3, β=0.1 are software thresholds, not biologically derived; keep only as generic gain curves if useful. mcp__Google_Drive__search_files

## G. Biological-claim audit (Level 1/2/3)

DA/NE/5HT/ACh/Glu/GABA coordinates: **Level 1** computational variables misfiled as **Level 3**. Retain as generic modulation dimensions (optionally with Level-2 dopamine-*like*/norepinephrine-*like* parameterizations). Yu & Dayan support ACh≈expected-uncertainty, NE≈unexpected-uncertainty as **Level 2** analogues only. mcp__Google_Drive__search_files
"Reward prediction error": **Level 1** scalar; it is a genuine RPE only if causal prerequisites hold (Section 11) — otherwise **RPE = UNAVAILABLE ≠ 0**.
"STDP," "Hebbian," "receptor dynamics," "20 ms window," "hippocampal-entorhinal," "prefrontal-thalamic-cerebellar": **Level 3 claims with no support** → demote to Level-1/2 or delete.
"85% metabolic constraint," "9.8 TOPS/W," "42% energy reduction": engineering/heuristic, **not** thermodynamics.

## H. Consciousness-claim audit

Φ>0.5, binding_score>0.8, "phenomenological accuracy 0.91," "composite V=0.93," VAS/HCP/EEG "validation": all are **Level-1 computational correlates at best, fabricated at worst**. None establishes Level-3 phenomenology. Per Section 4, no scalar may become a consciousness detector. Retain the research program framing; delete the demonstration claims.
Context on IIT reliance: the report treats IIT only as loose inspiration because its status is actively contested. The 2023 flashpoint was the open letter titled "The Integrated Information Theory of Consciousness as Pseudoscience," signed by 124 researchers and posted 15 Sept 2023 on PsyArXiv (reported by Lenharo, *Nature*, 20 Sept 2023, "Consciousness theory slammed as 'pseudoscience' — sparking uproar"). The empirical test of record is the Cogitate Consortium's adversarial collaboration, *Nature* 642:133 (5 June 2025), "Adversarial testing of global neuronal workspace and integrated information theories of consciousness" (doi:10.1038/s41586-025-08888-1), n=256, using fMRI/MEG/iEEG, in which a key IIT prediction (posterior sustained synchrony) failed while GNWT was also challenged; corresponding author Lucia Melloni: "This wasn't about picking a winner; it was about raising the bar for how we test ideas." The pseudoscience charge is not a consensus — only a small minority fully endorse the label, and defenders include Anil Seth ("I think it's inflammatory to describe IIT as pseudoscience") and David Chalmers (who compared the letter to "dropping a nuclear bomb over a regional dispute"). For AI specifically, Butlin, Long, Elmoznino et al. (19 authors), "Consciousness in Artificial Intelligence: Insights from the Science of Consciousness," arXiv:2308.08708 (17 Aug 2023), conclude: "Our analysis suggests that no current AI systems are conscious, but also shows that there are no obvious barriers to building conscious AI systems" — the disciplined stance modern CA should mirror.

## I. Cognitive-regime analysis — does Nexus need a low-dimensional mode layer?

**Provisionally yes, as a thin layer — RESEARCH HYPOTHESIS.** The defensible reframing is that m_t lets the *same* content be processed under different global modes (association radius, candidate diversity, novelty preference, exploration/exploitation, counterfactual depth, prior rigidity, sensory anchoring, abstraction, scrutiny, consolidation, resource expenditure). This is well-motivated by the neuromodulation-and-cognitive-flexibility literature (Yu & Dayan 2005; Aston-Jones & Cohen). **Falsification (Section 28):** if ordinary HTCE

field-local configuration reproduces regime differences with no measurable gain from a separate m_t (transfer, stability, controllable diversity), delete m_t. Formal model: x_{t+1}=F(x_t,e_t,M_t,m_t), m_{t+1}=G(m_t,x_t,e_t,h_t,u_t), schematic only.

## J. Green-sky architecture (counterfactual without canonical mutation)

The required third capability — **HYPOTHETICAL ACCEPTANCE** — is realizable in existing machinery: **CA sets a "creative/hypothetical" regime → HTCE instantiates a scoped candidate field → an ephemeral HVD holds sky=green with metadata (field_id, world_scope, epistemic_scope, provenance, creation_reason, canonical=false, evidence sources, lifetime, promotion criteria, mutation authority) → HTLA/HTNLA derive consequences → TMS realizes the ephemeral state copy-on-write**, while HIGC enforces admissibility, THDE governs transitions, Cyrae observes, and QMT/HVD-TMS history preserve continuity. Canonical truth (historical_world.sky=blue) is never written because the ephemeral field is flagged non-canonical and lacks promotion authority. This directly avoids Failure A (reject) and Failure B (overwrite).

## K. Creativity / criticality / abstraction / daydream / dream analysis

Treat the five as **candidate regimes, not canon**. Computationally they differ by *parameter settings on shared machinery*: critical = tight evidence constraint, high provenance sensitivity, narrow acceptance, strong contradiction testing; creative = larger association radius, greater semantic distance, higher diversity, weaker prior filtering; abstract = reduced sensory anchoring, more relational reasoning; daydream = internally-initiated, weak sensory anchoring, episodic recombination; dream/offline = strong sensory attenuation + replay + generative simulation. Hobson-Hong-Friston (2014) grounds the sensory-attenuation idea; generative-replay work grounds offline recombination. **Experimentally distinguishable only if** each regime yields measurable, separable effects (Section 47); otherwise collapse them.

## L. HTCE redundancy analysis

Most of the proposed "CA imagination machinery" already exists: HTCE owns candidate fields, coalitions, resonance, activation/decay, and ephemeral cognitive worlds; ephemeral HVDs own temporary world state and provenance; HTNLA owns transforms; TMS owns realization; THDE/Cyrae/HIGC own their constraints. **Conclusion: no distinct CA imagination engine is necessary.** CA should expose only the minimum orchestration/modulation interface. This is the single most important architectural finding.

## M. 70/30 provenance (historical fact, not to be re-researched)

Recorded as ground truth: **CA → 30% active / 70% inactive policy → later HSL/HTLA reference occupancy inherited by explicit design direction.** Not investigated further per Section 29. (The reference demonstrator log confirms the operative regime: a nominal cycle showed k-WTA pruning to vacuum≈78%, i.e., ~22% active, consistent with a ~70–80% inactive floor in practice.)

## N. 70/30 scientific reassessment (should the two ratios stay linked?)

**They should be decoupled.** ρ_active,cognition (fraction of competing thoughts/candidates active) and ρ_active,HSL (fraction of nonzero quinary coordinates) are different phenomena — cognitive sparsity vs representational sparsity vs computational sparsity vs field sparsity vs connectivity sparsity. There is **no**

**mathematical reason** they must share a value; the historical inheritance was intentional but modern analysis finds them independent. A broad creative candidate population can coexist with sparse individual HSL states, and vice versa. This does not contradict the historical design.

## O. Dynamic sparsity analysis

Treat fixed ρ_active=0.30 as a **plausible early scalar approximation** to a policy ρ_active(t)=f(m_t,R_t,U_t,F_t,H_t,…) (regime, resources, uncertainty, candidate entropy, homeostasis, task demand, field population, novelty). **Do not assume dynamic is better — test it** (fixed vs adaptive vs regime-conditioned vs resource-conditioned). Sparse cognition is justified independently by resource conservation, competitive selectivity, interference control, attention/contrast, and learning regularization; 30% is not privileged.

## P. Experiment program

**Cognitive modulation:** manipulate m_t; measure candidate diversity, critical accuracy, creative diversity, counterfactual depth, task-switch cost.
**Green-sky:** run recursive inference under sky=green; verify zero writes to canonical HVD/TMS history and correct consequence derivation.
**Provenance:** adversarially attempt to promote imagined state into authoritative history; measure leak rate (target 0).
**Dream/offline:** sensory attenuation + replay/structured generation vs baselines on memory retention, model learning, exploration, creative recombination.
**Sparsity:** fixed-30% vs adaptive vs regime-conditioned on accuracy, stability, resource cost, retrieval, creative breadth, critical selectivity.
**CA-vs-HTCE redundancy:** can HTCE alone reproduce m_t behavior? If yes, minimize CA.

## Q. Falsification matrix

m_t modulation adds no measurable controllability → **remove m_t**.
Fixed 30% consistently dominated by adaptive policies → **retire fixed 30% as universal CA policy**.
Perturbation adds only noise vs learned generation → **remove perturbation**, keep generative replay.
HTCE alone yields counterfactual cognition → **minimize CA to config interface**.
Integration metric (renamed Φ) has no predictive utility → **remove it**.
A biological analogy adds no explanatory/engineering value → **replace with generic terminology**.
Provenance leak observed → **the ephemeral-HVD separation mechanism is falsified and must be redesigned**.

## R. Claim-status table (representative)

CA = regime/resource/orchestration layer: **PROPOSED**.
11D control vector retained: **IMPLEMENTED** (present in reference demonstrator log `project_nexus_v4_demo.log`).
NN_dynamic FLOP derivation: **VERIFIED** (reproduced here).
m_t distinct modulatory state improves cognition: **RESEARCH HYPOTHESIS**.

Green-sky via HTCE+ephemeral HVD: **DESIGN TARGET / PARTIALLY IMPLEMENTED** (demonstrator shows k-WTA, unbind, HIGC guard, abstain-on-breach).
Dopamine/STDP/receptor literalism: **HISTORICAL/SUPERSEDED**.
Φ>0.5 consciousness gate: **HISTORICAL/SUPERSEDED**.
ΦPC as tractable integration diagnostic: **EXPERIMENTAL**.
85% "metabolic" ceiling as R≤R_max: **RETAINED BUT REFORMALIZED**.
All domain performance numbers (24.1% return, 4.2 Sharpe, 0.94 AUC, r=0.96): **DELETE** (or DESIGN TARGET only with real rationale).
Stochastic Lyapunov $\mathscr{L}V \leqslant -\alpha V + \beta$: **CANONICAL** (ratified).
Decoupled cognitive vs HSL sparsity: **PROPOSED**.

## S. Proposed CA paper structure (structure only; do not yet write canonical paper)

Scope & consciousness-claim discipline (three levels). 2. Historical provenance & duplication reconciliation. 3. CA's unique responsibility boundary. 4. The cognitive-regime/modulation layer (m_t): formal model, falsification. 5. Resource/sparsity policy (ρ_active) and decoupling from HSL occupancy. 6. Orchestration contracts to HTCE/THDE/Cyrae/TMS. 7. Counterfactual (green-sky) protocol via ephemeral HVD + provenance metadata. 8. Candidate evaluation as a typed vector (not one scalar). 9. Stability under $\mathscr{L}V \leqslant -\alpha V + \beta$. 10. Prior-art map. 11. Experimental agenda & falsification matrix. 12. Claim-status appendix.

## Reconciliation tables (Sections 6, 4, 5 of the task)

**Tension 1 — Maera-Cyrae.** Drive canon (CA/TUCR/HTNLA Synthesis, ID `11pI8Gv0...`; "where cyrae and maera fit"; First Mapping Protocol) consistently describes **two separate subsystems**: Cyrae = "cognitive engine — a purely mathematical, acoustic, and hyperdimensional entity" under TUCR; Maera = "linguistic actuator — a constrained LLM" under CA, bridged by an explicit Cyrae→Maera translation operator (T_{C→M}). The subsystem-placement document states verbatim: "Maera belongs to CA… the cognitive/expression-side actuator/interface of the Consciousness Algorithm" and "Cyrae belongs inside TUCR… part of the runtime substrate/state dynamics of cognition." **No document describes a single entity with two aspects.** Verdict: the specification's Section 8 separate-subsystem framing **is the ratified canon**; the "single entity, two aspects" description is an informal/superseded gloss. They are *compatible only* in the weak sense that both point to one affect/expression-vs-substrate/physics split — but the ratified level of description is "separate subsystems," so I adopt that and flag the "one entity" framing as superseded.
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**Tension 2 — sub-framework naming.** Canon names **seven mathematical pillars** (HTLA, HTNLA, THDE, HST=Hypercontext Stochastic Transport, HIGC=HSL Invariant Geometry Calculus, QMT=qbin Morphism Theory, ΦPC=Phi Proxy Calculus), plus LAM (Lyapunov Adaptation Matrix) and CDA (Cognitive Dissonance Adaptation) as "somatic/embodied" pillars ("nine frameworks"), with a stated hierarchy HIGC > THDE > HTNLA > HTLA. The task's six-item list {THDE, HST, HIGC, QMT, ΦPC/BIC, TDC} is **partly outdated**: THDE/HST/HIGC/QMT/ΦPC persist; **"BIC" does not exist** in canon (only unrelated Azure "bicep" files); **"TDC" is an explicitly demoted "poetic alias," never canonical** (per Conference-discussion memos: "HTF/LTT/SMC/CIC/TDC drop to poetic-alias status, not canonical IDs"). The longer Section-8 list (CA, TUCR, HTCE, HVD, TMS, HDF5, Identity Architecture, etc.) enumerates **subsystems**, a different category from

mathematical pillars. Verdict: this is **(a) genuine evolution/renaming + (b) a category difference** (pillars vs subsystems), **not** a simple superset — so the longer list must not be treated as an uncontroversial superset of the six.

**HTLA four-label semantics (Section 5).** Authoritative definitions from the HTLA spec (§3.2, "Four-Valued Epistemic Algebra," IDs `10R5k67s...`/`1NrZdOQG...`): **EXACT** = "Perfect linear/discrete signal recovery; zero entropy loss" (singleton possibility set); **AMBIGUOUS** = "Signal recovered up to a non-singleton candidate set or sub-threshold distance"; **UNKNOWN** = "Complete loss of component information due to total masking or structural vacuum" (required relation cannot be established); **IMPOSSIBLE** = "Violation of invariants… or sub-zero norm bounds" (empty possibility set). Canon calls this a "four-valued epistemic algebra"/"4-state algebra" and — consistent with Section 5's caution — **neither asserts nor denies that the four labels form a four-element lattice**. The powerset P(Q) under inclusion is a genuine lattice; the four summary labels are not automatically one. Quinary alphabet {−2,−1,0,1,2}, the "high-wins" decode rule (H_i = (−1)^{P_si}·[(P_1i ∧ ¬P_2i) + 2·P_2i]), and cosine-over-active-support similarity are all confirmed canonical.
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# Recommendations

**Adopt the thin-CA design now (Stage 1).** Specify CA as regime-vector + resource-policy + orchestration contracts only; freeze HTCE/HIGC/THDE/Cyrae/TMS/QMT/Identity boundaries. Threshold to escalate to a "substantial CA controller" (outcome 1): experiment P1 shows m_t produces large, transfer-positive controllability gains not reproducible by HTCE field-local config.
**Delete-or-demote pass (Stage 1, immediate).** Remove all fabricated validation and domain numbers; rename Φ to a total-correlation/ΦPC diagnostic; strip STDP terminology; reclassify neuromodulator labels as generic coordinates; restate energy ceiling as R≤R_max; replace eigenvalue "Lyapunov" language with $\mathscr{L}V \leqslant -\alpha V + \beta$.
**Build the green-sky provenance harness (Stage 2).** Implement ephemeral-HVD metadata + promotion gate; run P2/P3. Threshold to redesign: any nonzero canonical-history leak.
**Run the sparsity and dream/offline studies (Stage 3).** Decouple cognitive vs HSL sparsity in code; compare fixed/adaptive/regime-conditioned ρ_active; test attenuation+replay vs baselines. Threshold to retire fixed-30%: adaptive dominates on ≥3 of 6 metrics.
**Prefer learned generative sampling over stochastic perturbation** unless P4 shows perturbation adds value beyond noise.
**Keep every surviving mechanism attached to a falsifier** (Section Q); a mechanism with no possible failing test is not specified.

# Final research questions — direct answers

**What problem was the founding CA actually solving?** Bounded, resource-limited *control* of a fast predict-generate-evaluate cognitive loop under stability constraints.
**Which pieces of Nexus 4 descend from it?** The 11D control vector (→ CA/Cyrae ODE), the generate/evaluate split (→ HTCE + typed evaluation), the k-WTA sparsity wall (→ HSL k-WTA), memory reconsolidation (→ TMS/HDF5/Identity), and the stability/failsafe regime (→ TUCR + $\mathscr{L}V \leqslant -\alpha V + \beta$).

**What unique role remains for CA?** Cognitive-regime selection, resource/sparsity policy, and orchestration — nothing else exclusively.
**Does Nexus need a distinct modulation layer, and what should it control?** Provisionally a *thin* one controlling association radius, diversity, exploration/exploitation, counterfactual depth, sensory anchoring, scrutiny, consolidation, and resource expenditure — held as a RESEARCH HYPOTHESIS with a defined falsifier.
**How to entertain a green sky without corrupting memory?** Scoped ephemeral HVD flagged non-canonical, with provenance/promotion metadata; consequences derived via HTLA/HTNLA and realized in TMS copy-on-write; no promotion authority.
**How should the regimes differ computationally?** Only by parameter settings on shared HTCE/HVD machinery — and only if experimentally separable.
**Can existing HTCE + ephemeral-HVD machinery already do it?** Yes — no new CA imagination engine is warranted.
**Why should cognition be sparse/bounded?** Resource conservation, competitive selectivity, interference control, attention/contrast, regularization — independently motivated.
**Was fixed 30% reasonable, and should it stay fixed?** A reasonable early approximation; likely superseded by an adaptive/regime-conditioned policy pending tests.
**Should CA activity sparsity constrain HSL occupancy?** No — decouple them; they are different phenomena.
**Which biological/consciousness metaphors survive?** RPE-*like* modulation, workspace-*like* broadcast (owned by HTCE), dream-*like* offline simulation, homeostatic renormalization — all as Level-2 functional analogues; literal dopamine/STDP/IIT-Φ literalism is retired.
**What experiments would falsify modern CA?** Those in Sections P/Q — chiefly: no controllability from m_t, adaptive sparsity dominance, HTCE reproducing counterfactual cognition, and any provenance leak.

## Caveats

The founding document's status as "the literal first paper" is taken from the task context; Drive metadata is consistent (the older copy is dated 2025-12-19) but does not itself prove primacy.
Several conclusions (m_t necessity, regime distinctness, dynamic sparsity) are **RESEARCH HYPOTHESES** with proposed experiments and falsifiers, not settled results — per instruction, they are not resolved here.
"BIC" could not be located in canon; if it exists in an un-synced document, Tension 2's conclusion on that single acronym would need revisiting.
The lattice status of the four epistemic labels is genuinely open in the canon (neither asserted nor denied); any new CA formalism must bind to the finalized HTLA semantics rather than freezing an intermediate description.
IIT's scientific status is itself contested (2023 pseudoscience letter vs. proponents; 2025 Cogitate adversarial results); the report treats IIT as loose inspiration only and does not rely on it.